

Better or Worse For Your Workloads: Azure SQL vs. SQL in Fabric

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Why are you here?

Feature Comparison



Azure SQL Database



SQL Database in Fabric

Show of hands…



How many have seen/used Azure SQL Database?

How many have seen/used SQL in Fabric?



DEMO

Agenda

Architecture

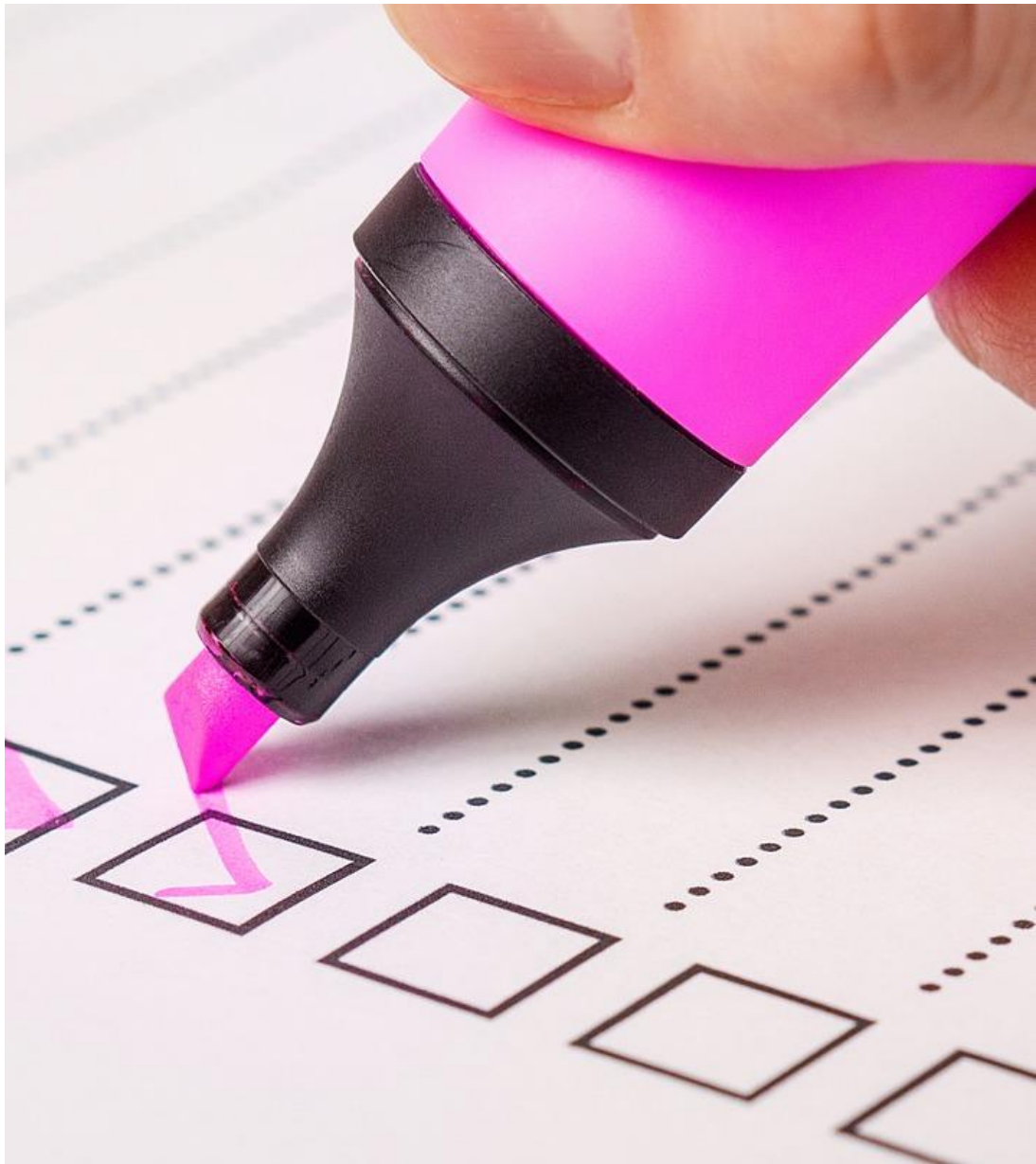
HA/DR

Costs & Pricing

Connectivity & Authentication

Q&A

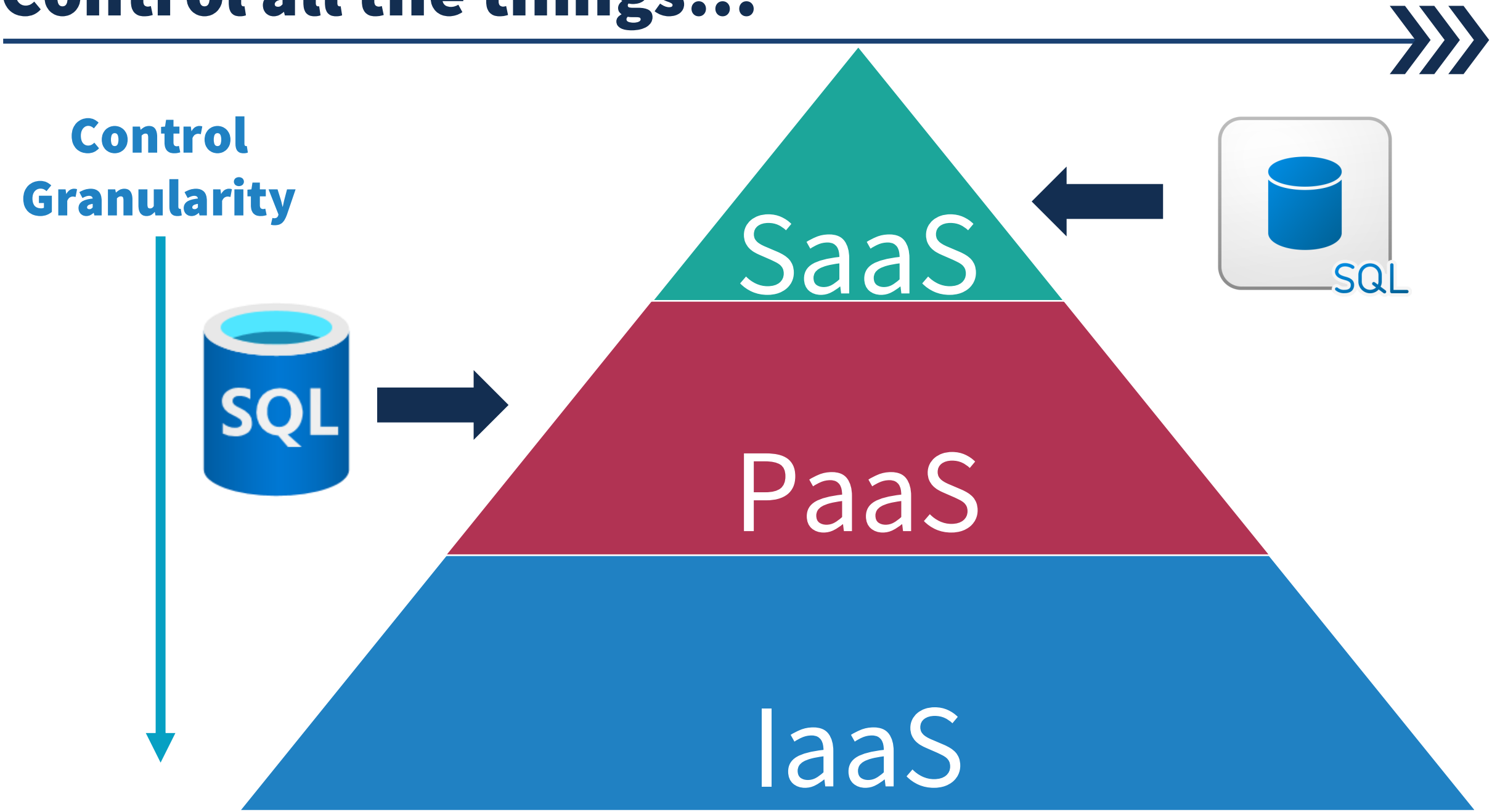
Summary





Architecture

Control all the things...



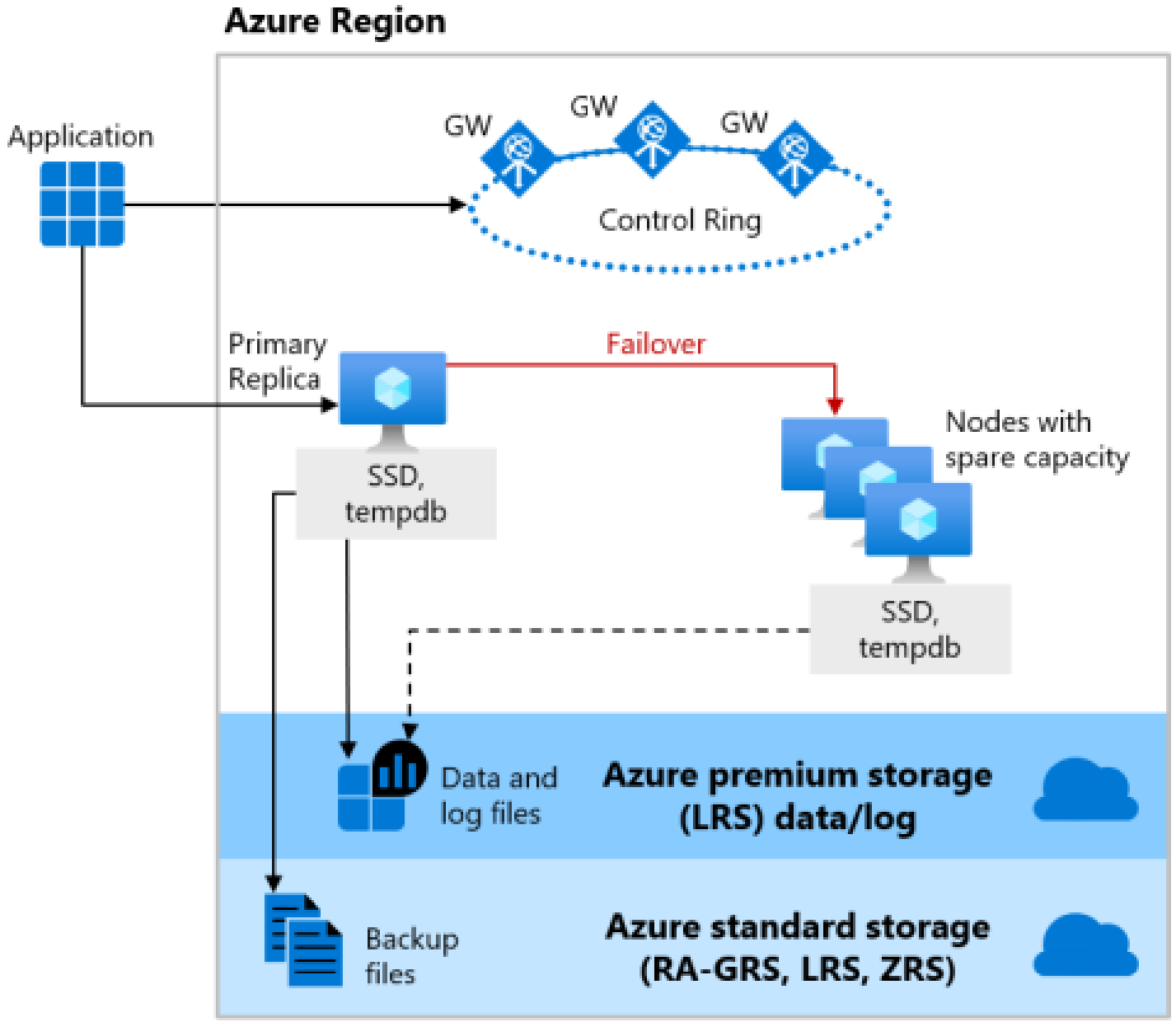
Azure SQL Database

General purpose

Network attached storage

Locally attached SSD for tempdb

Provisioned or Serverless



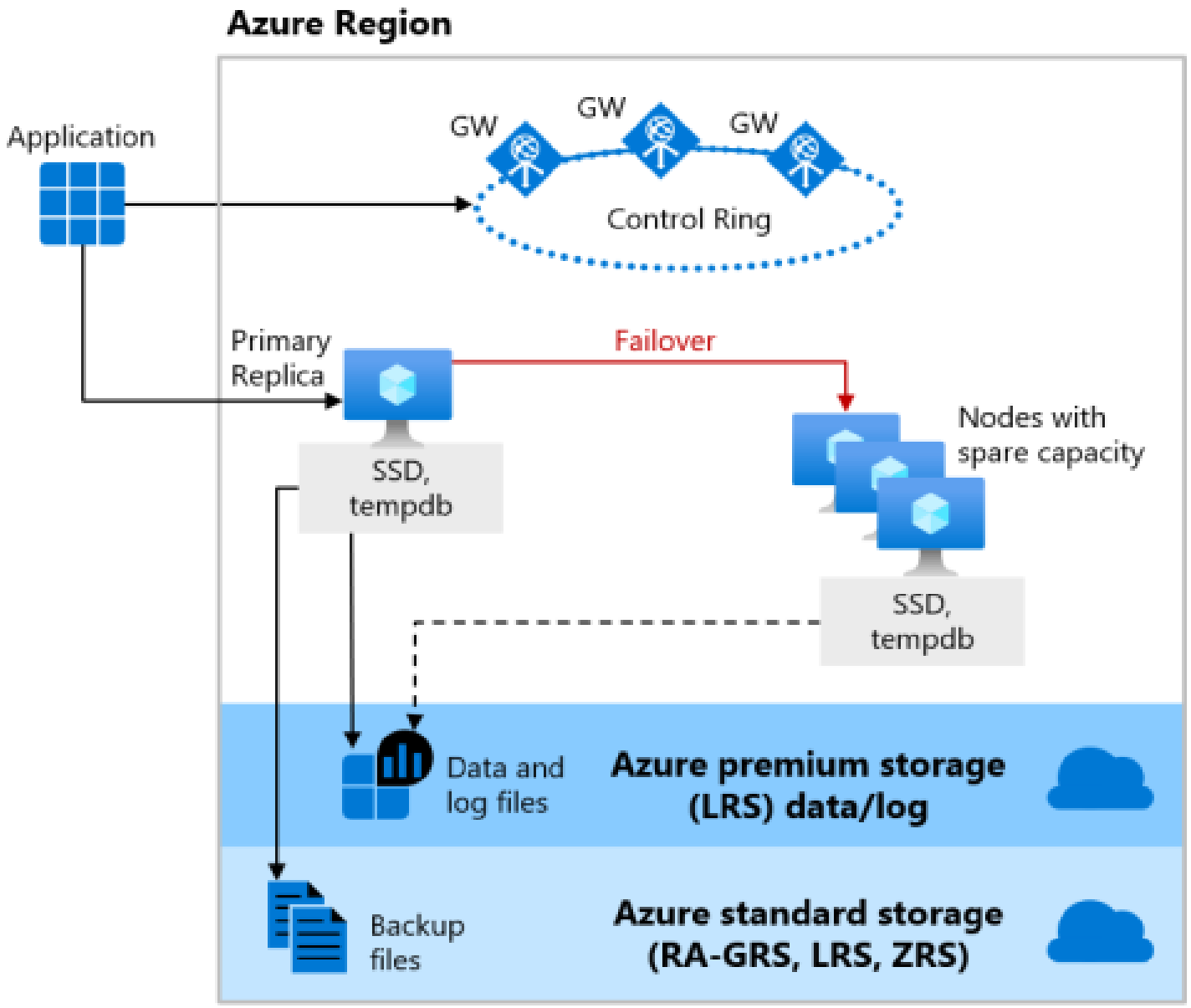


HA/DR

HA/DR

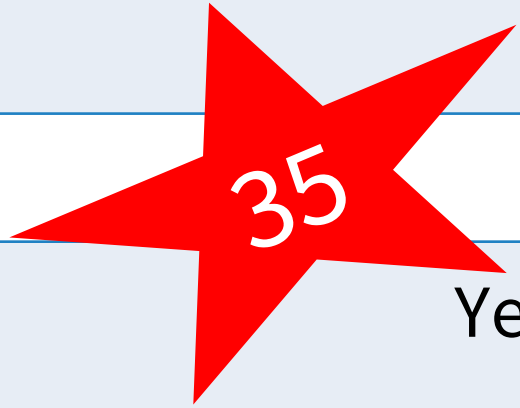
Stateless compute layer

Stateful data layer



HA/DR

Platform feature	Azure SQL Database	Fabric SQL database
Active geo-replication	Yes	No
Automatic backups	Yes	Yes
Availability zones	Yes	Yes, automatically configured
Failover groups	Yes	No
Geo-restore	Yes	No
Long-term retention (LTR)	Yes	No





Development Differences

Development Differences

Feature	Azure SQL Database	Fabric SQL database
Database compatibility	100 - 170	100-170
BULK INSERT statement	Yes, but just from Azure Blob storage as a source.	Yes, via OPENROWSET, with OneLake as data source.
OPENROWSET	Yes, from blob storage	Yes, with OPENROWSET BULK function (preview)
Collation - database collation	SQL_Latin1_General_CP1_CI_AS	SQL_Latin1_General_CP1_CI_AS
TDE	Yes	No
Always Encrypted	Yes	No
Columnstore indexes, clustered	Yes - For certain tiers	Yes, but the index must be created at the same time the table is created, or mirroring must be stopped.

Pricing

Azure SQL DB Pricing

vCore - ~$184 per provisioned core

vCore Serverless - $.5218/vCore-hour or $0.0001450/vCore-second

DTU - ~1.50 per DTU

Elastic pools are available for cost management

Reservations are available

Fabric Capacity

Defined as Capacity Units (CU)

Capacity units is a measurement of time for workloads

Capacity can service multiple workloads

F2 = 2 capacity units, 60 CUs in a 30 second block of time.

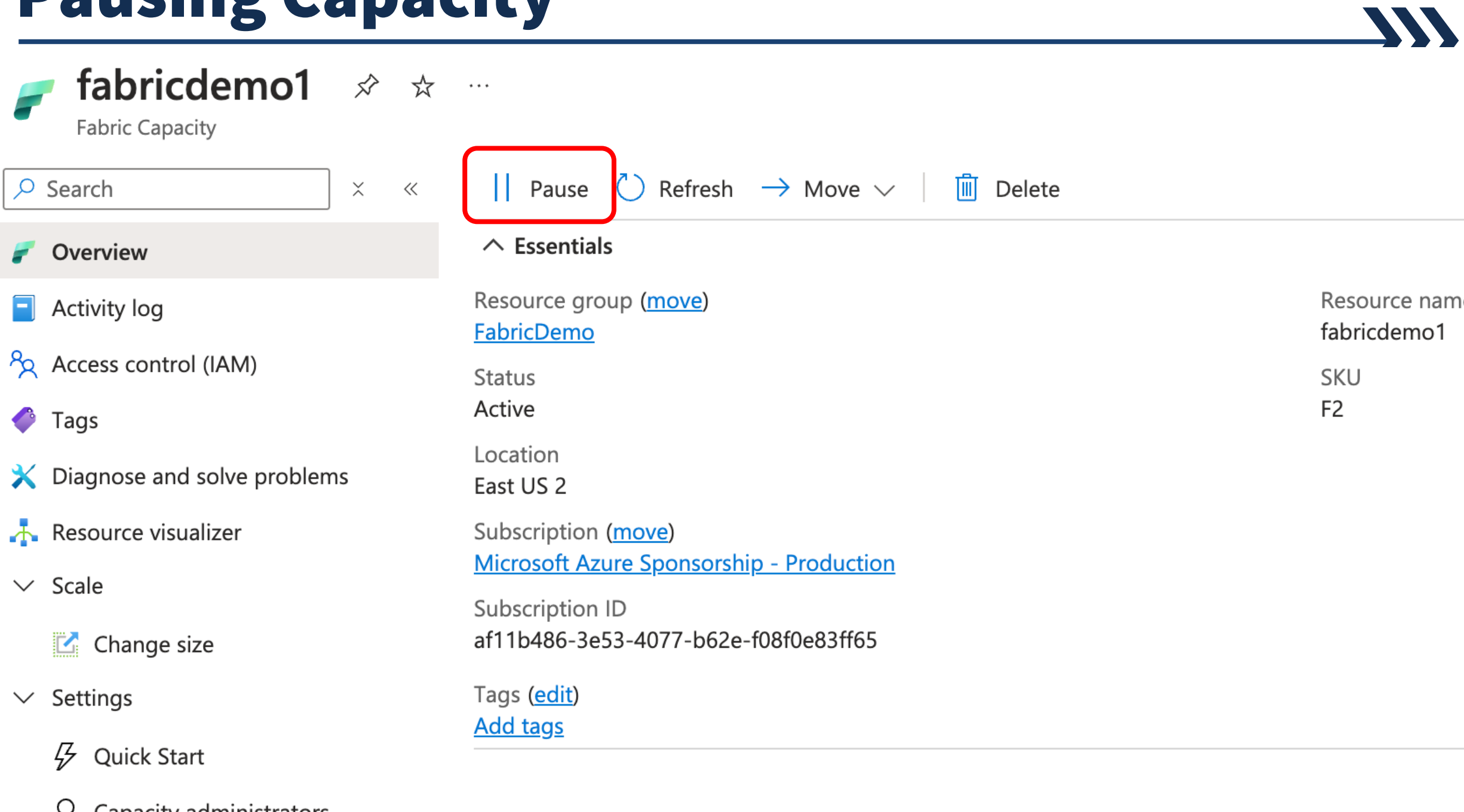
1CU = ~0.383 vCores. F64 = ~24 vCores

Reservations are available, elastic pools are not

Capacity Units Pricing – North Central

SKU	Cores	Capacity unit (CU)	Pay-as-you-go
F2	0.766	2	$262.80/month
F4	1.532	4	$525.60/month
F8	3.064	8	$1,051.20/month
F16	6.128	16	$2,102.40/month
F32	12.256	32	$4,204.80/month
F64	24.512	64	$8,409.60/month
F128	49.024	128	$16,819.20/month
F256	98.048	256	$33,638.40/month
F512	196.096	512	$67,276.80/month
F1024	392.192	1024	$134,553.60/month
F2048	784.384	2048	$269,107.20/month
F4096	1568.768	4096	$568,115.20/month
F8192	3137.536	8192	$1,076,428.80/month

Pausing Capacity



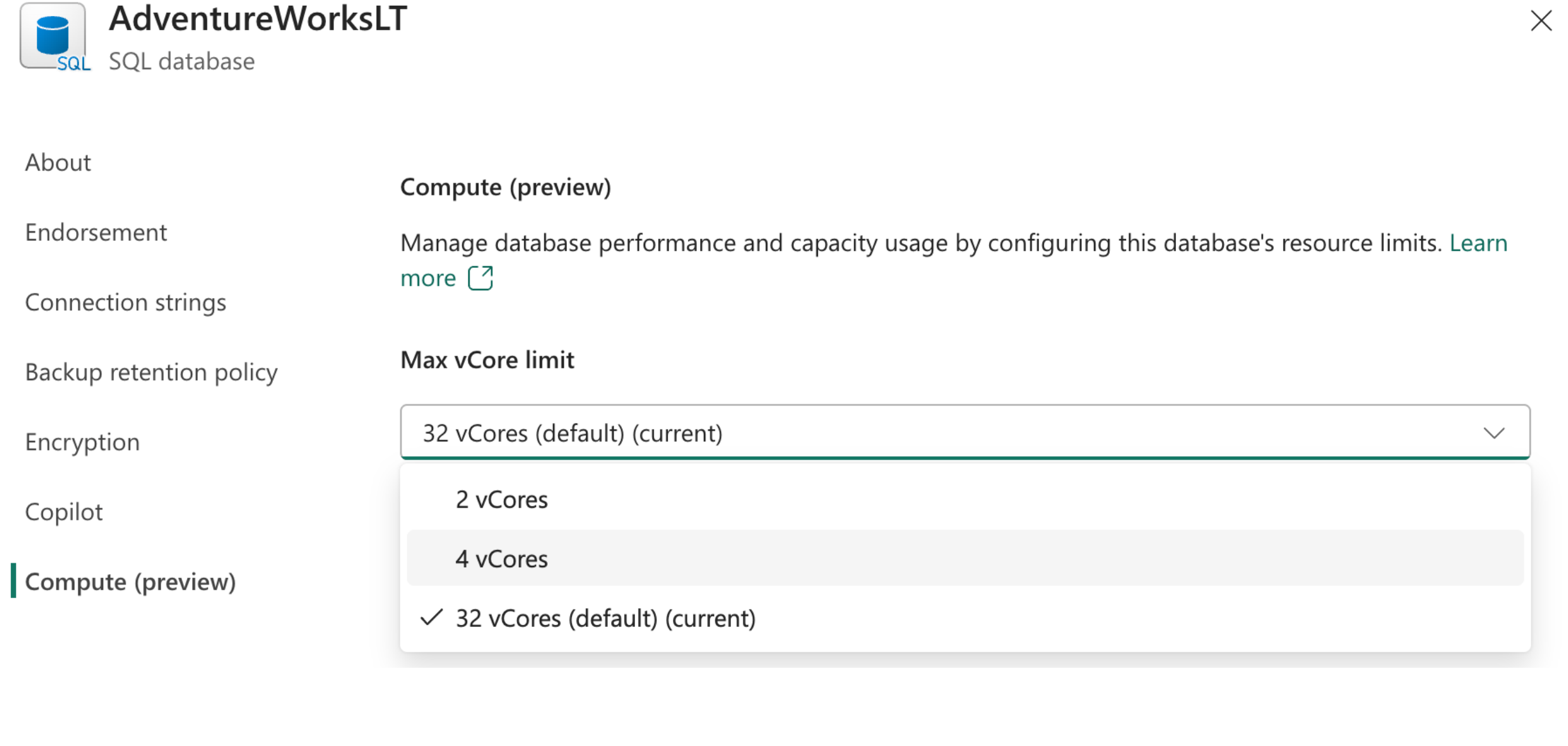
Fabric Capacity Planning

Primary Capacity— Day-to-day production workloads. Size appropriately to prevent EOM throttling.

Monitoring Capacity—Smaller capacity like an F4, and should just have your monitoring workspace(s) for the Fabric Capacity Metrics app.

Development Capacity—Used for development efforts, minimizes risks of development efforts impacting production workloads.

Throttle vCores





Connectivity & Authentication

Connectivity & Authentication

Azure SQL DB – SQL Authentication or Microsoft Entra

SQL Server Management Studio (SSMS), Azure Data Studio (ADS) , VS Code, Query Editor (Portal)

Private endpoints

Connectivity & Authentication

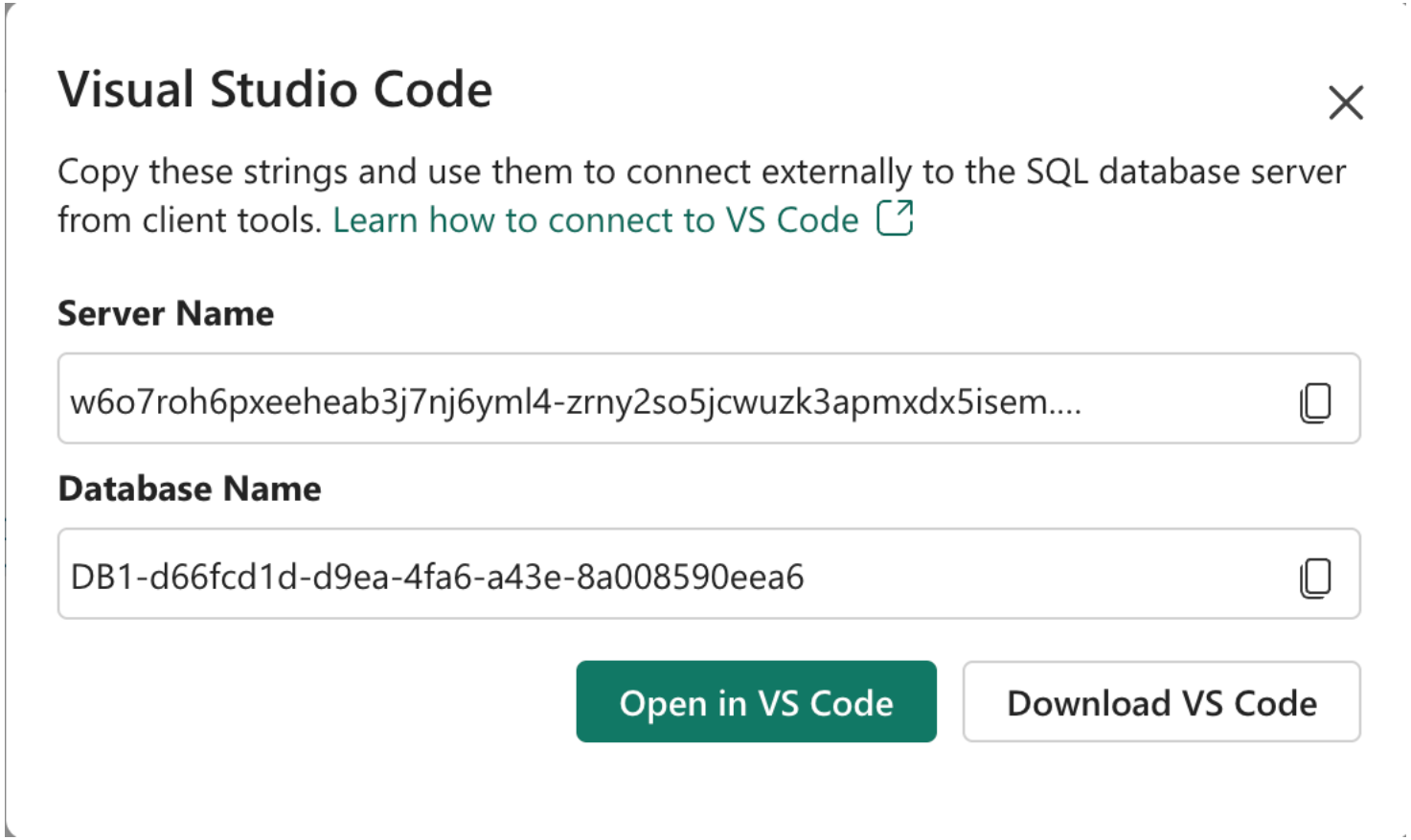
SQL Endpoint

Microsoft Entra Authentication only

VS Code / SSMS

Graphical Query Editor

Managed Private Endpoints



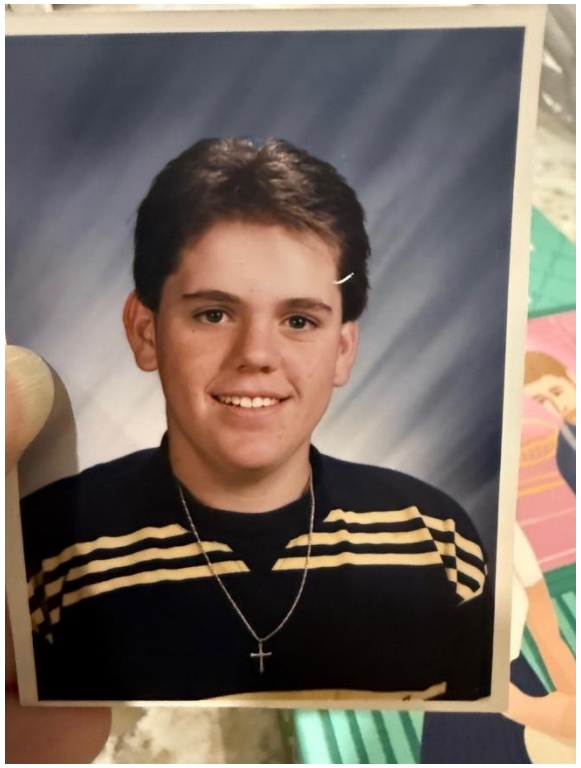
Core Feature Comparison

Feature	Azure SQL Database	SQL database in Fabric (preview)
Purchasing models	vCore, DTU	Provisioned Fabric capacity SKU
Compute tiers	Provisioned or serverless	Serverless
Elastic pools	Yes	No
Secure	Azure RBAC, SQL permissions	Fabric workspace roles and sharing permissions, SQL permissions
Authentication	Microsoft Entra ID authentication, SQL authentication	Microsoft Entra ID authentication
Database mirroring to Fabric OneLake	Yes, manually enabled	Yes, automatically enabled for all eligible tables
Cross-platform queries in Fabric	Yes, via mirroring to Fabric	Yes, via Fabric OneLake automatically

Maturity Comparison



Azure SQL



SQL in Fabric

Summary



Same basic architecture

SQL in Fabric cannot cross regional boundaries for DR

SaaS vs PaaS (less control)

Connectivity is different

In short…



It’s just SQL

So which database is more betterer?

Pick the right database for your workload...



References



Overview - https://learn.microsoft.com/en-us/fabric/database/sql/overview

Licensing - https://learn.microsoft.com/en-us/fabric/enterprise/licenses

Decision Guide - https://learn.microsoft.com/en-us/fabric/database/sql/decision-guide

Feature Comparison - https://learn.microsoft.com/en-us/fabric/database/sql/feature-comparison-sql-database-fabric



Questions?
Answers!

Your Next Speaker…

Monica Morehouse

Query Store: The simplest way to Troubleshooting Performance




THANK YOU
FOR YOUR
ATTENTION



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